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Review

Myosin phosphatase: Unexpected functions of a long-known enzyme☆

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ABSTRACT

Myosin phosphatase (MP) holoenzyme is a Ser/Thr specific enzyme, which is the member of protein phosphatase type 1 (PP1) family and composed of a PP1 catalytic subunit (PP1c/PPP1CB) and a myosin phosphatase targeting subunit (MYPT1/PPP1R12A). PP1c is required for the catalytic activity of the holoenzyme, while MYPT1 regulates MP through targeting the holoenzyme to its substrates. Above the well-characterized function of MP, as the major regulator of smooth muscle contractility mediating the dephosphorylation of 20 kDa myosin light chain, accumulating data support its role in other, non-contractile functions. In this review, we summarize the scaffold function of MP holoenzyme and its roles in processes such as cell cycle, development, gene expression regulation and neurotransmitter release. In particular, we highlight novel interacting proteins of MYPT1 and pathophysiological functions of MP relevant to tumorigenesis, insulin resistance and neurodegenerative disorders.

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1. Introduction

Reversible phosphorylation of proteins on serine (Ser), threonine (Thr) and tyrosine (Tyr) residues is one of the most common mechanisms by which coordinated regulation of cellular processes occurs [1]. The phosphorylation status of the cells is determined by the balance between the activities of protein kinases and protein phosphatases, therefore, the identification of the specific kinase(s) and phosphatase(s) implicated in the given cellular process is essential to uncover regulatory tools. Both types of enzyme are equally important in maintaining the physiologically relevant phosphorylation status of proteins, but they affect the phosphorylation signal differently. It is assumed that phosphatases may determine the time of the transduction of the signal to its intracellular target as well as the duration of the signal, while kinases have more influence over the strength of the signal [2]. There are hundreds of protein kinases and phosphatases as possible regulators of cellular events, however, these enzymes are interrelated too, as phosphatase mediation involves phosphorylation of phosphatase inhibitory and regulatory proteins, while dephosphorylation of kinases by phosphatases also contribute to their activation/inhibition. This

Abbreviations: AR, androgen receptor; CDK, cyclin dependent kinase; Chk1, Check point kinase 1; CLA, calyculin-A; CPI-17, C-kinase-activated protein phosphatase-1 inhibitor of 17 kDa; EGCG, epigallocatechin-gallate; eNOS, endothelial nitrogen monoxide synthase; iNOS, inducible nitrogen monoxide synthase; ERM, ezrin radixin moesin; FIH, factor inhibiting hypoxia inducible factor; HDAC7, histone deacetylase 7; GBPI, gastrointestinal-and brain-specific PP1 inhibitor; ILK, integrin linked kinase; IP3K, inositolphosphatae 3 kinase; IR, insulin receptor; IRS1, insulin receptor substrate 1; KEPI, kinase enhanced PP1 inhibitor; LATS1, large tumor suppressor kinase 1; LPS, lipopolysaccharide; 67LR, 67 kDa laminin receptor; M20/21, small subunit of myosin phosphatase of 20/21 kDa; MLC20, 20 kDa regulatory light chain of myosin II; MLCK, MLC20 kinase; MLCP, myosin light chain phosphatase; MP, myosin phosphatase; M-RIP/p116RIP, myosin phosphatase-Rho interacting protein; microRNA-30d, miR-30d; MYPT1, myosin phosphatase target subunit 1; Nkx2.5, NK2 homeobox 5; OA, okadaic acid; Par-4, prostate apoptosis response protein; NUAK1, NUAK family SnF1-like kinase-1; PDK, pyruvate dehydrogenase kinase; PHI-1 and -2, phosphatase holoenzyme inhibitor-1 and -2; PI3K, phosphatidylinositide 3 kinase; PKA, cyclic AMP-dependent protein kinases/protein kinase A; PKC, protein kinase C; PKGIα, cGMP-dependent protein kinase Iα/protein kinase G; PLK1, polo-like kinase 1; PP1c, protein phosphatase 1 (PP1) catalytic subunit; pRb, retinoblastoma protein; PP2Ac, protein phosphatase-2A (PP2A) catalytic subunit; PP2B, protein phosphatase-2B; PRMT5, protein arginine methyltransferase 5; ROCK, Rho A associated kinase; SIAH2, Seven in absentia homolog 2; SMTNL1, smoothelin like 1 protein; SNARE, soluble *N*-ethylmaleimide sensitive factor attachment protein receptor; SNAP-25, synaptosomal-associated protein of 25 kDa; SKM, skeletal muscle; VSMC, vascular smooth muscle cell; ZIPK, zipper-interacting protein kinase

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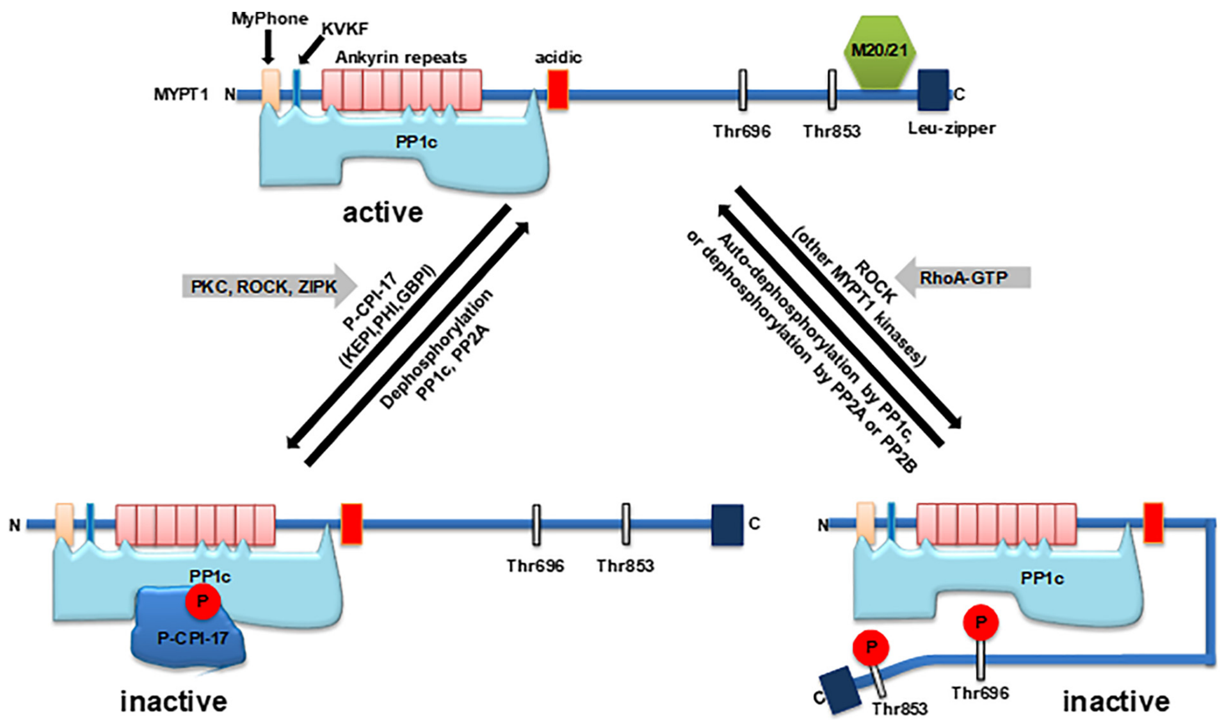

Fig. 1. The structure and regulation of MP by phosphorylation. The upper part of the figure depicts the major structural regions of MYPT1 and its interaction with PP1c and M20/21. The lower part represents the inhibition of MP by direct phosphorylation of its MYPT1 regulatory subunit at Thr696 and Thr853 (right). A scheme for inhibition by phosphorylated CPI-17 is also shown. Possible dephosphorylating phosphatases are also indicated under the reverse arrows. See text for more details for the phosphorylation dependent regulation of MP activity.

review mainly focuses on the regulatory possibilities of the phosphoserine/threonine (P-Ser/Thr) specific protein phosphatases with a special emphasis on the cellular roles of myosin phosphatase (MP). The major aims are to reveal primarily the novel, non-contractile functions of this enzyme as it has received also considerable attention for the past few years. Even though, this introduction may not disregard a brief retrospective summary of studies on MP in relation to myosin phosphorylation and smooth muscle contraction as well as the structure and regulation of MP.

Besides the definite physiological significance, smooth muscle has been the tissue in which numerous signaling pathways (Ca^{2+}-signal, NO/cGMP effects, etc.) are studied in connection with the changes in contractile features. Even in protein phosphorylation history, the regulation of smooth muscle contraction via phosphorylation-dephosphorylation of the 20 kDa regulatory light chain of myosin II (MLC20) was investigated soon after the recognition that the active form of glycogen phosphorylase required covalently bound phosphate (reviewed in [1]). The first studies were focused on the MLC20 kinase (MLCK) [3], and the phosphatase was thought to ensure only the reversibility of the modification by dephosphorylation. The role of MLC20 phosphatase (MLCP) in smooth muscle contraction was clearly revealed when okadaic acid (OA), a potent membrane permeable phosphatase inhibitor was discovered. OA induced muscle contraction parallel with MLC20 phosphorylation in skinned gizzard preparation even in the absence of Ca^{2+} [4]. These data indicated that phosphorylation of MLC20 may occur via the possible involvement of Ca^{2+}-independent kinase(s) distinct from the classical Ca^{2+}-calmodulin dependent MLC20 kinase (MLCK) if the phosphatase is suppressed in the system. At about the same time Ca^{2+}-senzitization of smooth muscle contraction in G-protein coupled pathways was also demonstrated with the possible involvement of MLCP (reviewed in [5]). These findings directed attention to the structure and regulation of MLCP, and although there were attempts before to isolate this enzyme, many distinct forms had been reported even without a consensus in the type of the phosphatase.

The myosin phosphatase (MP) was isolated first from chicken gizzard [6,7] and later from pig bladder [8]. These independent groups confirmed that the MP/MLCP holoenzyme consists of three subunits: a protein phosphatase-1 catalytic subunit (PP1c) of about 38 kDa; a large myosin binding subunit of 110–133 kDa (later defined as myosin phosphatase target subunit-1, MYPT1); and a smaller subunit of about 20/21 kDa (M20/21). It is to note that the myosin phosphatase is named by different acronyms such as MP, MLCP or PP1M. We use MP in further description of the structure and the function of this enzyme in the cellular processes and the name PP1cβ/δ (PPP1CB) for the catalytic subunit and MYPT1 (PPP1R12A) for the 130–133 kDa major regulatory subunit. The latter protein is also termed as M110, myosin binding subunit (MBS), and M130/M133 in earlier publications.

1.1. The structure and regulation of MP

The interactions of the subunits in the trimeric holoenzyme of MP have been a great interest for the past three decades and identified by distinct binding techniques and structural analysis. It was apparent that PP1c was responsible for the catalytic activity and it interacted with MYPT1, which plays a targeting role in the dephosphorylation process as it also binds to myosin directing PP1c toward the phosphorylated residues in MLC20. The M20/21 subunit binds to the C-terminal region of MYPT1 and had no direct influence on phosphatase activity. While MYPT1 is ubiquitously expressed in mammalian cells, M20/21 is found, besides smooth muscle, mainly in cardiac muscle. The physiologic role of M20/21 is largely unrevealed yet, but the data have gained so far suggest that it increases Rho-A associated kinase (ROCK) dependent inhibitory phosphorylation of MYPT2 (the heart specific MYPT isoform) in cardiomyocytes resulting in suppression of MP activity [9]. It has also been shown that M20/21 regulates microtubule dynamics [10]. It was established that out of three major PP1c isoforms (α, β/δ, γ1) expressed in mammalian cells MYPT1 exclusively interacted with PP1cβ/δ [11] and this specificity was determined by residues in PP1c located at the central region. The structure of MYPT1 and its isoforms has been described in several reviews [12,13] together with other MYPT family proteins such as MYPT2 [14], MBS85, MYPT3 and TIMAP [15]. Our present review focuses on the physiological roles of the ubiquitously expressed MYPT1 complexed with PP1cβ/δ in distinct cell types.

The schematic structure of MYPT1 in association with PP1c and M20/21 is shown in Fig. 1. MYPT1 includes an RVxF (consensus sequence RxxQV/I/LK/RxY/W) PP1c-binding motif ($K^{35}VKF^{38}$) which appears to be essential for the interaction of MYPT1 with PP1c [16]. Additional binding sites are: (i) a region N-terminal to the RVxF motif (residues 10–17 in MYPT1 termed the myosin phosphatase N-terminal element or MyPhoNE); (ii) repeats 1, 5, 6 and 7 of the ankyrin repeat region [17]; and (iii) region(s) (not identified) between residues 301 to 511 close to the central region [16].

It has been an attractive idea that the interaction of PP1c and MYPT1 might be regulated by the phosphorylation of MYPT1 in the N-terminal region. Thr34 at the proximity of the PP1c-binding motif ($^{35}KVK^{38}F$) was a good candidate to disrupt subunit interactions after phosphorylation. However, Thr34 phosphorylated by PKC did not either strengthen or weaken the interaction of MYPT1 with PP1c [16]. Recently, Hu et al. [18] identified Ser20 as a phosphorylation site for CHK1 (Check point kinase 1) and this phosphorylation event is claimed to be essential to the interaction of MYPT1 with PP1c as well as to the dephosphorylation of PLK1 by MP. On the other hand, phosphorylation of Ser20 decreases the stability of MYPT1 rendering it more susceptible for proteasome degradation. It is unclear yet how these opposite effects, i.e. promoting activity of MP toward PLK1 and at the same time decreasing activity by MYPT1 degradation, are balanced to reach beneficial physiological responses.

A binding site for PP1c in the C-terminal half of MYPT1 has not been identified. Nevertheless, this MYPT1 region plays an important role in the phosphorylation-dependent inhibition of MP activity. MYPT1 includes several consensus sites for a number of protein kinases and among these phosphorylation at Thr696 and/or Thr853 (numbering is for the human isoform) results in inhibition of MP activity (Fig. 1). ROCK was first identified as the kinase for the phosphorylation of both Thr696 and Thr853 [19], but later several other kinases including zipper-interacting protein kinase (ZIPK), integrin-linked kinase (ILK), myotonic dystrophy kinase, Raf-1 [20] and p21-activated protein kinases were also shown to act on Thr696 (reviewed in [12,13,21] and see references therein). Using Thr696Ala and Thr853Ala mutants of MYPT1 it was also established that phosphorylation of either Thr696 or Thr853 resulted in inhibition of PP1c in a reconstituted PP1c-MYPT1 complex in vitro [22]. The current hypothesis is that phosphorylation of these sites induces such structural movements in MYPT1 which allows the inhibitory phosphorylation peptide to interact with the substrate binding groove and the catalytic center of PP1c (Fig. 1). These suggested conformational changes might be helped by the hypothesized, partly “unstructured” nature of the C-terminal regions of MYPT1. However, around the inhibitory phosphorylation site (Thr696) the protein is structured. It was demonstrated in a study by NMR [23] of a MYPT1 peptide (658–714) including Thr696 that the average structure of this peptide is relatively extended and consists of three helices. The Thr696 is located at the end of the middle helix and the loops connecting the helices are thought to be quite flexible, therefore, they may allow the necessary movements upon phosphorylation leading to inhibition. The sequence around P-Thr696 or P-Thr853 are similar to that of P-Ser19 in MLC20 [17] therefore these MYPT1 regions may act as substrate analogues which are dephosphorylated by PP1c at a very slow rate compared to P-Ser19 of MLC20.

Thr696 and Thr853 inhibitory phosphorylation sites are preceded by Ser695 and Ser852, residues in MYPT1 which are phosphorylated by cyclic AMP- and cGMP-dependent protein kinases (PKA and PKG). First, it was demonstrated that PKA or PKG dependent phosphorylation of Ser695 counteracted with phosphorylation of Thr696 protecting MP from ROCK- or other MYPT1 kinase dependent inhibition [24]. These phosphorylation assays were further developed to identify cross-talk between the PKA and ROCK-dependent phosphorylation events [25,26] and suggested that PKA phosphorylation at Ser852 also abolish subsequent phosphorylation at Thr853, while an interesting finding was that PKA is able to phosphorylate simultaneously all four residues (Ser695, Thr696, Ser852, Thr853) but this MYPT1 form is not inhibitory on PP1c.

Besides the direct inhibition of MP activity by phosphorylation of MYPT1, MP activity could also be suppressed by inhibitory proteins in an indirect manner (Fig. 1). The best known MP inhibitor is the C-kinase-dependent phosphatase inhibitor of 17 kDa (CPI-17) which is active following phosphorylation at Thr38 by protein kinase C (PKC). Other kinases (ROCK, ZIPK) may also be involved in Thr38 phosphorylation (reviewed in [27]). CPI-17, unlike other PP1 inhibitors, such as inibitor-1 and -2, can inhibit both PP1c and MP (PP1c-MYPT1 complex) with high effectiveness, and although it does not contain a canonical PP1c binding motif, its interaction with both PP1c and MYPT1 is apparent. Other CPI-17 family members: phosphatase holoenzyme inhibitor-1 and -2 (PHI-1 and -2), kinase enhanced PP1 inhibitor (KEPI) and gastrointestinal-and brain-specific PP1 inhibitor (GBPI) include PP1c-binding RVxF motif and an inhibitory phosphorylation sequence very similar to that of in CPI-17 [27]. KEPI also inhibits both PP1c and MP [28] and its phosphorylation and inhibitory action on MP in THP-1 and in MCF-7 cells is also considered [29].

An important question with respect to the regulation of MP which phosphatase(s) can relieve MP from the inhibition by phosphorylated MYPT1 or CPI-17. Isolated PP1c, protein phosphatase-2A (PP2A) catalytic subunit (PP2Ac) and protein phosphatase-2B (PP2B, also termed calcineurin) dephosphorylated the phospho-Thr696 and -Thr853 in MYPT1 (Fig. 1). It was also established that cyclosporin A, a calcineurin/PP2B inhibitor, increased the phosphorylation level of both Thr696 and Thr853 in MYPT1 as well as influenced the barrier function in endothelial cells suggesting that PP2B plays an important role in the dephosphorylation of these inhibitory sites in cellular systems [30]. Similarly, calyculin-A in a PP2A inhibitory concentration resulted in significant elevation in the phosphorylation of Thr696 of MYPT1 and suppression of the barrier function in endothelial cells implicating PP2A in $MYPT1^{pThr696}$ dephosphorylation [31]. These are indications for the roles of PP2B and PP2A in dephosphorylation of MYPT1 and activation of MP and imply a specific interplay between the distinct types of phosphatase in cellular regulation. In recent works 67 kDa laminin receptor (67LR) mediated activation of MP was reported in melanoma cell lines [32] and in endothelial cells [31] by epigallocatechin-gallate (EGCG), a green tea ingredient. In an EGCG/67LR → PKA → PP2A → MP pathway activation of PP2A (containing Bδ subunit) by PKA phosphorylation results in enhanced dephosphorylation of both $MYPT1^{pThr696}$ and $CPI\text{-}17^{pThr38}$ thereby activation of MP.

The above results point to the possibility that PP2A may also be a $CPI\text{-}17^{pThr38}$ phosphatase. However, a recent model has been worked out stating that the inhibitory action of CPI-17 ^{pThr38} on MP could be due to “unfair competition” with the substrates, implying that the inhibitor binds tightly excluding interactions of PP1c (or MP) with the substrates and on the other hand interaction of $CPI\text{-}17^{pThr38}$ with a dephosphorylating phosphatase [33]. In addition, $CPI\text{-}17^{pThr38}$ is a “bad” substrate for MP, it is dephosphorylated slowly and this also contributes to a sustained inhibitory action. Nonetheless, the dephosphorylation of $CPI\text{-}17^{pThr38}$ by PP1c is considered to be sufficient and necessary for the reactivation of MP. Based on this suggested model, a similar “autodephosphorylation” mechanism may play a role in the reactivation of MP from inhibition caused by phosphorylation of MYPT1 at Thr696 and/or Thr853. The MYPT1 subunit of MP is subject to posttranslational modifications other than phosphorylation also and these events may affect the physiological role of this enzyme. MYPT1 is a substrate for the asparaginyl hydroxylase FIH [factor inhibiting HIF (hypoxia inducible factor)] and it is hydroxylated in the ankyrin repeat region in cultured cells and in vitro using purified proteins [34]. The physiplogical consequence of this modification with respect to MP activity is unclear. It is suggested that MYPT1 serving as an alternative substrate of HIF may influence HIF/hypoxia signaling.

It has also been shown that MYPT1 is subject to reversible methylation by histone lysine methyltransferase SETD7 and histone demethylase LSD1 [35] at Lys442. As methylation stabilizes MYPT1, increased demethylation activity which is quite common in malignant cells results in degradation of MYPT1 by the ubiquitin-proteasome pathway, favoring retinoblastoma protein phosphorylation and cell cycle progression of cancer cells. These data suggest that MYPT1 protein stability is regulated by its methylation status influencing the phosphorylation level of protein(s) dephosphorylated by MP involving the targeting function of MYPT1.

Molecular interaction between Seven in absentia homolog 2 (SIAH2) and MYPT1 has been shown to be necessary for the proteasomal degradation of MYPT1 in mammalian cells [36]. This interaction involved the substrate binding domain of SIAH2 (116–324) and a central region of MYPT1 (445–632) which included a degenerate consensus SIAH-binding motif of RLAYVAP (493–499). SIAH2 is an E3-ligase capable of directing proteins for degradation by the ubiquitin–-proteasome system. These data confirm that the cellular level of MYPT1 and thereby the activity of MP is regulated by proteolytic degradation. In this regard it is interesting that MYPT1 is also subject to proteolytic cleavage by caspase-3 during apoptosis of cells [37]. Caspase-3 cleavage at the C-terminal region (Asp884) results in loss of the myosin binding ability of MYPT1 concomitant with increased inhibitory phosphorylation at Thr696 and Thr853. As a consequence of cleavage of the myosin binding region, MYPT1 dissociated from actin stress fibers and this delocalization together with enhanced inhibitory phosphorylation led to hyperphosphorylation of myosin-II.

1.2. Interactions of MYPT1

The unique structure of MYPT1 provides platforms for multiple interaction with numerous proteins and other biological macromolecules (Fig. 2 and Table S1). It is also important that which region(s) of MYPT1 is/are responsible for the binding of these interactive partners (Fig. 2). First, the binding of the major substrate, P-myosin, was probed and although these experiments led to controversial results from different groups the consensus is that P-myosin interacts with both the N-terminal and C-terminal regions in a way that P-MLC20 bind at the second half of the ankyrin repeat region while the heavy chain associates with the C-terminal part [12]. A similar binding pattern is established for the retinoblastoma protein (pRb) [38] and also hypothesized for the interaction of MYPT1 with the endothelial nitric oxide synthase (eNOS) [31].

Binding of the other MYPT1 partners are more region selective (reviewed in [12,13]). Adducin, Tau, MAP2, protein methyltransferase 5 (PRMT5) and SNAP-25 all bind to the N-terminal region and they are phosphorylated by ROCK and dephosphorylated by MP. Smoothelin-like protein-1 (SMTLN1) as well as calcineurin (PP2B), a MYPT1 phosphatase, binds preferentially also to the N-terminal half which contains a specific PIxIT motif for its binding. In contrast, GTP-RhoA, moesin (and ezrin), interleukin-16 precursor protein [39], prostate apoptosis response (Par-4) protein [40], myosin phosphatase-Rho interacting protein (M-RIP/p116RIP) and cGMP-dependent kinase (cGKIα) bind to the C-terminus and within this group interleukin-16 precursor protein, Par-4, M-RIP and cGKIα specifically interact with the leucine-zipper motifs. The diversity of these interactions predicts the role of MP in a great deal of cellular processes and the significance of these protein-protein interactions will be highlighted in the upcoming sections (Table 1).

2. Diverse functions of myosin phosphatase

2.1. Basic role of myosin phosphatase in cytoskeletal processes

MP has an established role in the control of actomyosin contractility via dephosphorylation of MLC20 but it also contributes to the regulation of cell migration [41] or cell adhesion [42]. MP also plays a role in the regulation of cytoskeletal organization by dephosphorylation of the ERM proteins. The ERM family consists of three structurally related proteins, ezrin, radixin, and moesin, which function as linkers between transmembrane proteins and the underlying cytoskeleton, and fulfill many cellular functions in cell adhesion, migration, or proliferation. Inactive, ERM proteins reside in the cytoplasm in a closed conformation, while phosphorylation at a conserved threonine residue (Thr567 in ezrin, Thr564 in radixin, and Thr558 in moesin) by ROCK or PKC induces the formation of an open and active conformation enabling them to bind to their target molecules [43,44]. Several reports suggest that MP is responsible for dephosphorylation of ERM in different cell types [44–46]. Direct interaction between C-terminal part of MYPT1 (rat sequence) and the N-terminal domain of ezrin and moesin was identified, and MP was shown to dephosphorylate the C-terminal part

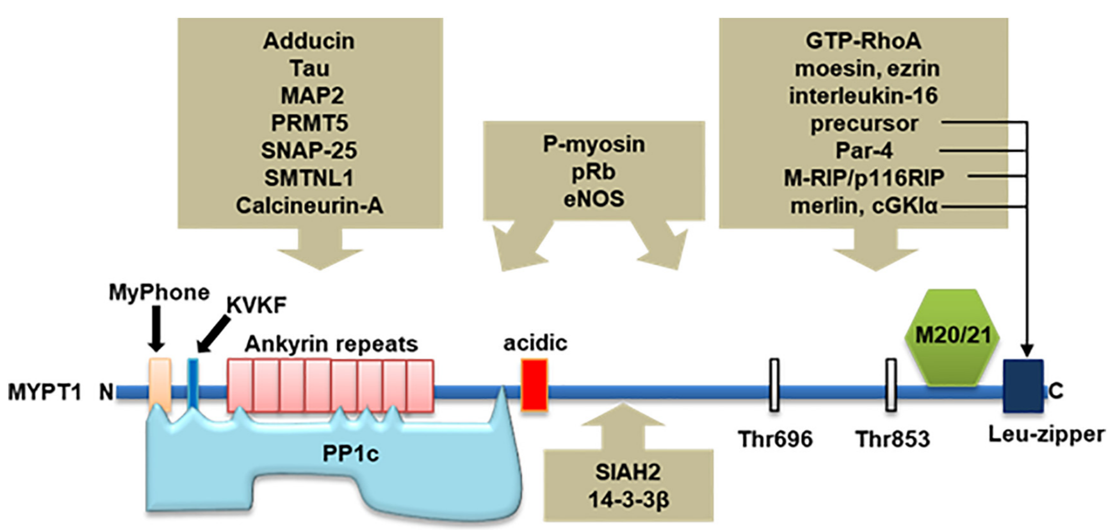

Fig. 2. Protein-protein interactions of MYPT1. Besides PP1c a few major interacting partners of MYPT1 are shown classified as binding to the N-terminal region (left panel above the MYPT1 scheme), to both the N-terminal and C-terminal region (middle panel) and to the C-terminal regions (right panel). Arrows in the right panel indicate proteins which interact with the leucine zipper motifs. Proteins binding to the central region are indicated below the MYPT1 scheme.

Table 1
Substrates of myosin phosphatase.

MP substrate	Tissue/cell type	Regulated P-site	Protein kinase	Citation
MLC20	VSM	Thr18	MLCK/ROK	[5,6]
ERM				
Ezrin	HPAEC	Thr567	ROK	[44]
Radixin	HPAEC	Thr564 (direct dephosphorylation was not proven)	ROK	[44]
Moesin	HPAEC, MDCK	Thr558	ROK	[43,44]
Adducin	MDCK (only interaction was proven)	Not identified	ROK	[48]
Tau	Brain	Thr245/Thr377/Ser409 (direct dephosphorylation was not proven)	ROK	[45]
pRb	THP-1	Thr826	CDK4	[38]
	SBC-5	Thr807/811	CDK4	[35]
eNOS	BPAEC	Thr497	PKC/ROK	[31]
SNAP25	Mouse cortex, B50	Thr138	ROK, PKA, PKC	[74,75]
Syntaxin	Mouse cortex, B50	Ser14	ROK	[74]
Synapsin	Mouse cortex, B50	Ser9	ROK	[74]
PRMT5	Human hepatocarcinoma cell	Thr80	ROK	[54]
IRS1	Skeletal muscle, L6	Ser522	AKT?	[125]
Polo-like kinase 1	SW962, HeLa	Thr210	Aurora A	[62]
HDAC7	Thymocytes	Ser155	PKD1	[90]
		Ser318		
		Ser448		

of moesin phosphorylated by ROCK in vitro [44]. In another study, selective association of ERM proteins and MP subunits were revealed by coimmunoprecipitation: ezrin and moesin directly interacted with PP1cδ, while radixin was bound to MYPT1 only [45]. MYPT1 was found to colocalize with moesin at membrane ruffling area in MDCK epithelial cells suggesting that MP may play a role in the regulation of moesin phosphorylation in vivo [44]. Dephosphorylation of ezrin by MP was demonstrated by MYPT1-depletion or CPI-17 overexpression experiments in different cell lines [47]. MP-catalyzed dephosphorylation of ERM proteins has a protective effect against LPS-induced endothelial permeability [46]. Overexpression of CPI-17 in human melanomas showed Ras activation and tumorigenic transformation via inhibition of MP and subsequent activation of ERM proteins [47]. MP together with ROCK was identified as a regulator of the phosphorylation state of adducin, which mediates the interaction of adducin with F-actin [48].

The microtubule associated proteins, Tau and MAP2 were also found to interact with both ROCK and MYPT1, and MP was shown to dephosphorylate ROCK-phosphorylated Tau in vitro in PC12 pheochromocytoma cells proposing that MP may also regulate microtubule dynamics in neuroblastoma cells [49].

2.2. Role of MP in the control of cell proliferation and cell division

2.2.1. Dephosphorylation of retinoblastoma protein

MP contributes to the control of cell proliferation by distinct mechanisms [50]. One remarkable cell cycle-related interactor of MYPT1 is retinoblastoma protein (pRb), a tumor suppressor that is dis-regulated in many types of cancer. pRb regulates cell cycle and proliferation by repressing transcription of E2F-mediated genes. Phosphorylation by cyclin-dependent kinases inactivates pRb and promotes cell cycle progression. PP1 or PP2A are responsible for dephosphorylation and activation of pRb [51,52]. MYPT1 was identified as a possible PP1-targeting subunit acting on phosphorylated pRb in leukemic cells [38]. The interaction of MYPT1 and pRb was confirmed at cellular level, and surface plasmon resonance (SPR) studies revealed that the C-terminal fragment of pRb binds to both the N- and C-terminal part of MYPT1. The MYPT1-PP1c complex enhanced the catalytic activity toward phosphorylated pRb compared to PP1c alone, supporting the targeting function of MYPT1 during pRb dephosphorylation. Inhibition of MP either by CLA or by the overexpression of the endogenous MP inhibitory protein KEPI [29] results in an increased phosphorylation of pRb-Thr826, and this may contribute to the chemoresistance of leukemic cells caused by phosphatase inhibition. In another study, overexpression of MYPT1 resulted in a decrease in the phosphorylation of pRb at Ser807/811 residues, suggesting that MP may modulate pRb activity via dephosphorylation at multiple sites [35]. MP-catalyzed dephosphorylation of pRb was proposed to play a role in the regulation of androgen receptor (AR) which plays a pivotal role in the development and progression of prostate cancer [53]. Analysis of clinical datasets showed downregulation of MYPT1 and upregulation of KEPI in human prostate cancer cells. Restoring MP activity in prostate cancer cells by KEPI depletion was able to repress aberrant AR activity by the following mechanism: activated MP dephosphorylates pRb at Ser807/811 residues causing repression of E2F1 mediated cell cycle progression and subsequent AR phosphorylation by CDKs and a significant inhibition of AR target gene expression leading to reduced proliferative and migratory capacity. In addition to dephosphorylation-dependent regulation, MP was also implicated in the control of protein expression of pRb via dephosphorylation and inhibition of arginine methyltransferase PRMT5 ([54], see in details later in this review). These data imply that MP may be involved in the regulation of cell cycle progression by controlling the activity of pRb at multiple levels.

2.2.2. Dephosphorylation and activation of merlin

The NF2 tumor suppressor gene product, merlin was also reported to be dephosphorylated and activated by MP [55]. Merlin has similar domain organization as the ERM family proteins, but it is regulated in an opposing manner, that is, phosphorylated merlin is inactive, while the dephosphorylated form is active and functions in contact inhibition and growth suppression. Merlin inhibits cell proliferation via multiple mechanism, it initiates anti-mitogenic signaling and suppresses oncogenic gene expression [56,57]. MP was identified as the decisive phosphatase that renders merlin active in its closed conformation by dephosphorylation at Ser518 residue. MYPT1 binds through its leucine zipper domain to the amino-terminal part of the alpha-helical coiled coil region of merlin, and L339 side chain of merlin was shown to be critical for the interaction. Depletion of MYPT1 elevates merlin phosphorylation and supports Ras activation. CPI-17, the specific inhibitor of MP, is upregulated in many cancer cells, and causes the loss of merlin function in tumor suppression [55]. ILK was identified as the specific kinase responsible for the inhibition of MP in cancer cells. By direct phosphorylation of MYPT1 ILK promotes the inactivation of merlin and concomitant suppression of the Hippo tumor suppressor pathway which leads to the activation of YAP/TAZ/TEAD-mediated transcription of growth-promoting genes. [58]. Recently, nuclear association of MYPT1 and merlin in cardiomyocytes was reported. MP was shown to be activated in response to oxidative stress mediating the dephosphorylation and activation of merlin, thereby modulating Hippo signaling in cardiomyocytes [59]. Interestingly, curcumin – a natural compound with anti-inflammatory and anti-tumorigenic effects – seems to assist the activation of merlin by dephosphorylation-dependent activation of MP [60].

2.2.3. Regulation of the dephosphorylation of polo-like kinase 1 by MP during mitosis

MYPT1 is specifically phosphorylated during the course of mitosis by proline-directed kinases, and phospho-MYPT1 was shown to associate with the mitotic polo-like kinase 1 (PLK1) on centromeres and on kinetochores [61,62]. PLK1 is a Ser/Thr kinase that controls a variety of mitotic events including G2/M transition, maturation of centrosomes, mitotic spindle assembly, sister chromatid segregation, mitotic exit, and cytokinesis. Silencing of MYPT1 results in an elevated phosphorylation of PLK1 at its activation site, Thr210, suggesting that MP may antagonize the function of PLK1 during mitosis. Accordingly, simultaneous reduction of MYPT1 and PLK1 protein level rescue cells from mitotic arrest caused by inhibition of PLK1 alone [62]. Studies in the last few years have demonstrated that via spatial and temporal coordination of PLK1 activity MYPT1 contributes to the control of mitotic progression. For example, MYPT1-depleted cells enter M phase more quickly, implicating MP in G2 checkpoint regulation [63]. Loss of MYPT1 also results in precocious sister chromatid segregation [64] as well as the reversal of the step-wise increase in the kinetochore-microtubule assembly stability between prometaphase and metaphase [65]. Binding and activity of MP toward PLK1 is modulated by multiple phosphorylation sites and several kinases (Fig. 3). Proline-directed kinases, CDK1 (cdc2) and MAPK were shown to phosphorylate Ser432, Ser473 and Ser601 residues in MYPT1 during mitosis. Phosphorylation of Ser473 by Cyclin A/Cdk1 complex constitutes a recognition site in MYPT1 for the Polo-box domain of PLK1, which is crucial for the association of MYPT1 and PLK1 [62,65]. An adaptor protein, optineurin was shown to be required too for bringing CDK1 in the vicinity of MYPT1 and stimulate CDK1-mediated MYPT1-Ser473 phosphorylation and subsequent PLK1 dephosphorylation during mitosis [66]. It was suggested that phosphorylation of MYPT1-Ser473 alone is not sufficient for proper activation, but Ser445 phosphorylation is also required to stimulate MP activity toward PLK1. The latter phosphorylation is mediated by LATS1, another mitotic kinase [63]. Upon cell detachment, NUAK1 phosphorylates MYPT1 at Ser472 along with Ser445 and Ser910 residues, which was shown to trigger binding of MYPT1 to 14–3-3 protein thereby inhibiting the activity of MP [39,42]. In addition, the phosphorylation of these side chains, most notably Ser472, may also interfere with the “priming” phosphorylation of MYPT1 at the PLK1 docking site, Ser473. Consistent with this, NUAK1 activation leads to increased PLK1 phosphorylation, presumably by weakening the association between MYPT1 and its substrate [67]. No discussion is given by the authors on the contradictory effect of Ser445 phosphorylation on MP activity. NUAK is activated by Liver kinase 1 (LKB1), and it was reported that shRNA-mediated inhibition of LKB1 strengthens the association between MYPT1 and PLK1, providing further evidence for the negative regulatory role of NUAK1 in MYPT1-mediated dephosphorylation of PLK1 [68]. It is concluded that LKB1 stabilizes the MYPT1-PP1c- PLK1 complex, however this is not fully consistent with the observed experimental data. In addition, it was proposed recently that CHK1-mediated phosphorylation of MYPT1 at Ser20 is also required to target MP to PLK1 [18].

2.3. Novel role of MP in neurotransmitter release

Recent studies have shown that MP is localized in neurons and it is also involved in the mediation of neuronal processes. Neuronal myosin phosphatase target subunit 1 was first identified from brain cDNA library screened with a cDNA fragment of rat MYPT1 [69]. The global mapping of rat brain revealed that expression of MYPT1 was detectable in each brain region but showed higher concentration and greater enzyme activity in cortex, olfactory bulb, striatum, hippocampus, and diencephalon with a concentration of 30–78 nmol/g total protein [70].

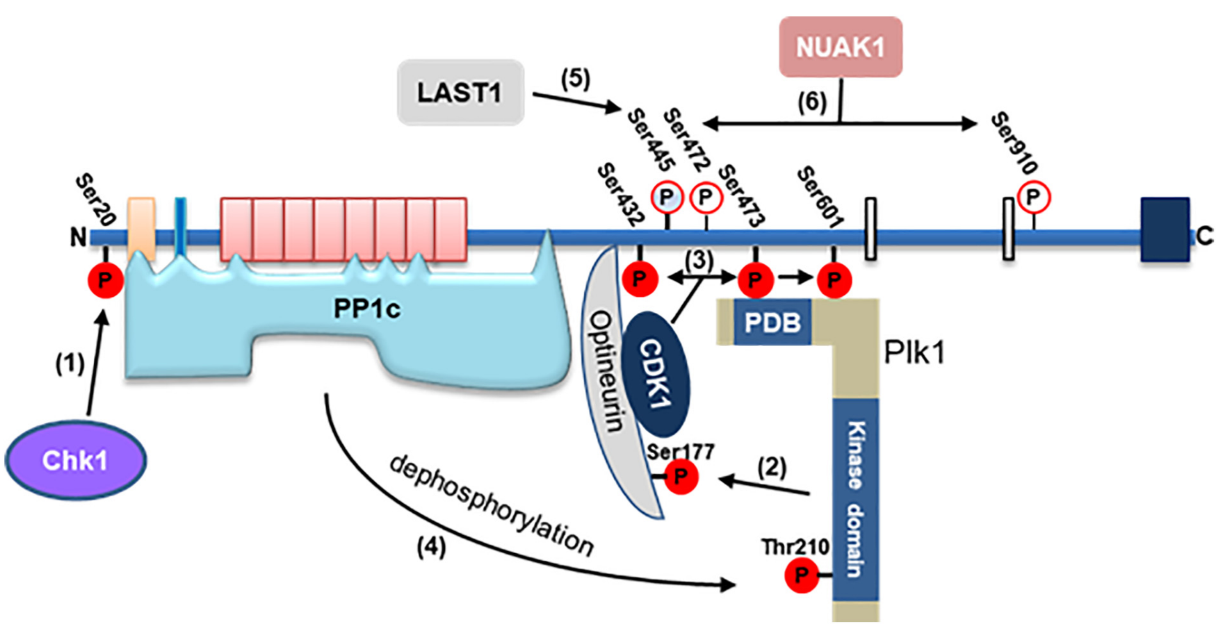

Fig. 3. Regulation of myosin phosphatase in mitosis.
Modulation of MP activity toward Plk1 by phosphorylation and protein-protein interaction. (1) Check point kinase 1 (Chk1) phosphorylates MYPT1 at Ser20 which was suggested to be necessary for MYPT1-PP1cδ interaction. (2) Optineurin is phosphorylated by PLK1 at Ser177, stimulating the binding of optineurin to both MYPT1 and CDK1, and (3) allowing CDK1 to phosphorylate MYPT1-Ser473 (as well as Ser432 and Ser601). This phosphorylation creates a polo-box domain binding pattern in MYPT1, required for the interaction between MYPT1 and Plk1, and (4) subsequently targets MP toward Thr210 in Plk1. (5) Phosphorylation of MYPT1-Ser445 by LATS1 was also suggested to promote MP-catalyzed dephosphorylation of Plk1. (6) NUAK1 has a negative role in the regulation of MP-mediated activation of Plk1 by phosphorylation of Ser445, Ser472, and Ser910 residues in MYPT1 and thereby weakening the association between MYPT1 and Plk1. *Plk1*: *polo-like kinase-1*; *PBD*: *polo-box domain*; *CDK1*: *cyclin-dependent kinase 1*; *Chk1*: *checkpoint kinase 1*; *NUAK1*: *NUAK family SnF1-like kinase-1 LATS1*: *large tumor suppressor kinase 1*.

These values were comparable to the MYPT1 concentrations found in C2C12 myoblast cells (40–48 nmol/g total protein) [71]. The presence of MYPT1 was also observed in cortical synaptosomes and in primary neuronal cells. The complex of MYPT1 and PP1cδ was colocalized and coprecipitated with synaptophysin, a presynaptic marker protein [70]. Mass spectrometry analysis identified several additional MYPT1-bound synaptosomal proteins (Table 1) such as synapsin-I, calcineurin-A subunit, Ca^{2+}-calmodulin dependent kinase II and the SNARE (soluble *N*-ethylmaleimide sensitive factor attachment protein receptor) complex elements such as syntaxin, and the synaptosomal-associated protein of 25 kDa (SNAP-25). The phosphorylation of syntaxin-1 at Ser14 by casein kinase-2 was already described [72] and it was suggested that the phosphorylation of synapsin-I by PKA at Ser9 modulates neurotransmitter release through docking actin to synaptic vesicles [73]. SNAP-25 was also identified as a substrate of ROCK that acts on the Thr138 phosphorylation site of SNAP-25. Dephosphorylation of all the previously mentioned ROCK-specific sites was catalyzed by MP [74]. The application of PP1-specific inhibitors on cortical synaptosome preparations resulted in the suppression of the depolarization-induced exocytosis and neurotransmitter release supporting the previous findings [74]. Moreover, the transduction of MP inhibitor KEPI protein into synaptosomes resulted in an increase of SNAP-25 phosphorylation and a decrease in the extent of neurotransmitter release and the interaction between SNAP-25 and syntaxin increased with decreasing phosphorylation level of SNAP-25 on Thr138 [75] suggesting that MP positively regulates neurotransmitter release through the initiation of the formation of SNARE complex (Fig. 4).

2.4. Functions of MP during development

Embryonic development is the subsequent sequence of regional specification, morphogenesis, cell differentiation and growth. Regional specification involves the action of cytoplasmic components that are located within the fertilized egg and it does not generate functional differentiated cells. Morphogenesis relates to the formation of three-dimensional shape and it involves the synchronized movements of cells creating the three germ layers of the early embryo. Cell differentiation relates specifically to the formation of functional cells. Growth mostly occurs through cell division but also through changes of cell size and the deposition of extracellular materials [76].

The role of MP in embryogenesis has been already suggested since *MYPT1*-deletion caused embryonal lethality at stage e7.5 days [77]. The majority of publication documented MP action on development through the modulation of the actomyosin complex that generates forces driving tissue morphogenesis [78]. Live imaging studies demonstrated that in adult Drosophila ovary during egg formation MP via its MYPT1 subunit (flw Drosophila form) is the key regulator of the morphogenesis of follicle cells. MP was required for the initiation and periodicity of basal myosin oscillation that creates egg chamber elongation [79].

Pregnancy and the implantation of the zygote is one of the key step in embryogenesis and during this process tissues coordinately adapts in order to accommodate the developing fetus and then prepare for delivery. One of the inhibitory molecule of MP, namely smoothelin like protein 1 (SMTNL1) was identified as a target of PKA and PKG in the course of smooth muscle contraction [80]. It was characterized as a protein with a dual function: it serves as the negative regulator of MP in the cytoplasm but upon phosphorylation by PKA/PKG at the Ser301 phosphorylation site, SMTNL1 translocates to the nucleus and acts as a cofactor of progesterone receptor. During pregnancy and aging the expression of SMTNL1 increased and it was correlated with MYPT1 expression [81]. MYPT1 expression increased drastically during development and sexual maturation and it was sex-dependent since it was 30% higher in males than females in VSM and skeletal muscle.

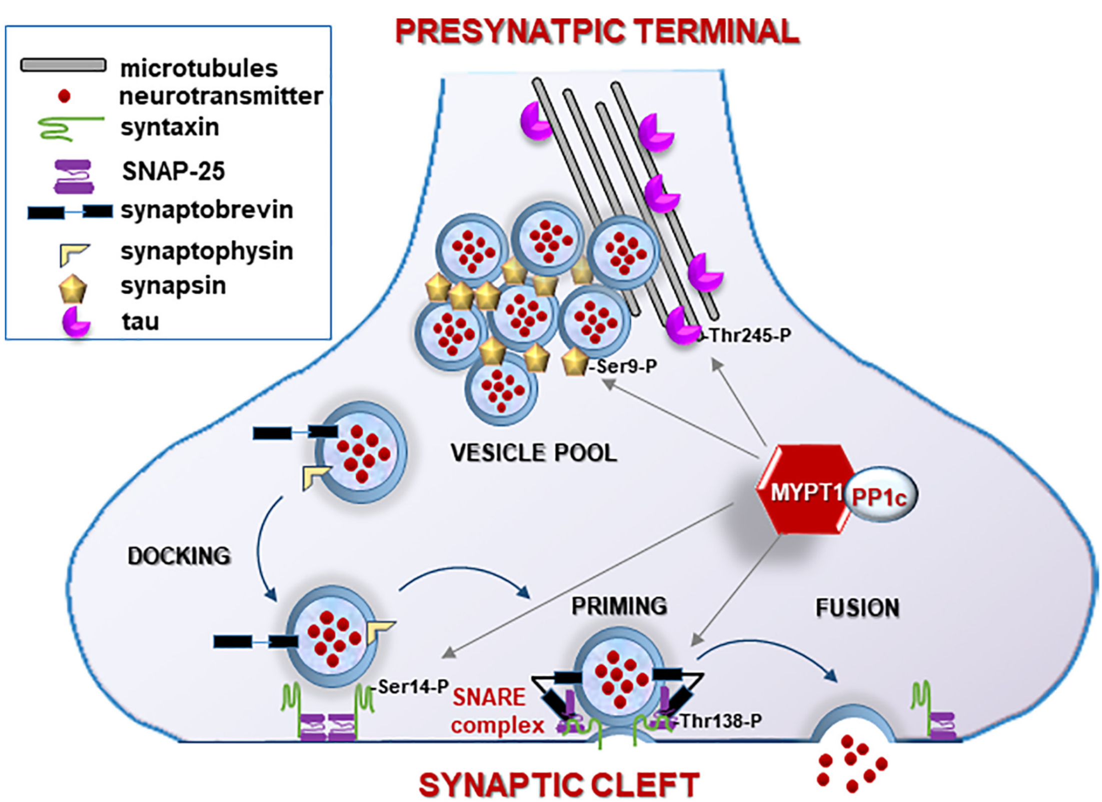

Fig. 4. MP regulates neurotransmitter release.
MP dephosphorylates syntaxin at Ser14 and SNAP-25 at Thr138 which initiates the priming of synaptic vesicles. The phosphorylation level at Ser9 of synapsin modulates its affinity to the synaptic reserve of vesicles regulating their transport to the active zone. Tau phosphorylation regulates microtubules assembly and vesicle trafficking.

However, *SMTNL1*-deletion caused a reciprocal age-dependent pattern of expression of MYPT1 as well as a change in MP activity. Pregnancy as well as SMTNL1 deletion significantly increased the expression of MYPT1 in smooth muscle suggesting a pregnancy-related Ca^{2+}-desensitization and relaxation regulating blood pressure. [82]. SMTNL1 regulated the expression of MYPT1 at transcriptional level in various sex-hormone sensitive tissues as well as in smooth and skeletal muscle where it induced an isotype switch of muscle fibers to a more glycolytic phenotype causing global metabolic shift toward insulin resistance [83]. Based on these findings it is clear that SMTNL1 regulates metabolic, contractile and reproductive properties, and known mechanisms are through the regulation of MP at the transcriptional level and also by inhibiting its enzymatic activity in development and pregnancy.

During development the tight coordination of adjacent tissues are required. Such coordination was characterized in the motor neuron positioning in Zebrafish during axonogenesis. The cell bodies of neurons in developing vertebral column and the axons in the cord exiting to the surrounding tissues shift relative to each other because of their differential growth rate. MP was identified as a regulator of the motor neuron adjustment in their position and it was believed to be a conserved feature in development. *Mypt1* mutants exhibited motor neuron defects and it was found to modulate caudal primary axon branching through a common, muscle intrinsic mechanism [84].

MP was also affected in human embryonic stem cells. Administration of H89 PKA inhibitor resulted in an increased cell viability, suppressed cell blebbing and colony formation of the dissociated stem cells, however, the pluripotency of hESCs and their ability to differentiate were not affected. PKA inhibition likely led to a decrease in ROCK activity and caused changes in the adherent properties of hESCs were accompanied by a decrease in MYPT1 and MLC20 phosphorylation [85]. Therefore, both the cell differentiation and the embryogenetic contributions of MP should be further investigated to clarify its role in developmental stages.

2.5. Regulation of gene expression by MP

MYPT1 was also found in the nuclear and microsomal fraction of HepG2 cells [86], and in the nucleus of neuronal cells [70] proving its role beyond the regulation of cytoskeletal elements and contractility. One of the unconventional nuclear substrates of MP is histone deacetylase 7 (HDAC7), which belongs to the class IIa HDACs. This is a member of a transcriptional complex involved in gene expression regulation and it acts as a repressor of numerous genes such as *Nur77* that involved in T-lymphocyte differentiation [87]. HDAC7 translocates upon its phosphorylation on the Ser155, Ser318 and Ser448 sites by protein kinase D to the cytoplasm where it associates with 14–3-3 proteins [88,89]. HDAC7 attaches in the cytoplasm to MP holoenzyme [90], which dephosphorylates the phospho-Ser residues of HDAC7 and promotes its nuclear redistribution [91]. Subsequent attenuation of Nur77 (an orphan nuclear receptor also known as NR4A) expression inhibits apoptosis of thymocytes implicating a regulatory role of MYPT1 in the survival of thymocytes (Fig. 5).

Another regulatory action of MP on gene expression is via the regulation of the subcellular localization of Nkx2.5 transcription factor. Nkx2.5 is one of the earliest markers of cardiomyogenic differentiation and its activity is required for cardiac differentiation [92]. It regulates many targets through direct interactions with other transcription factors including GATA4, Tbx5 and MEF2C [93]. MYPT1 was identified as a putative cofactor of Nkx2.5 and their interaction has been verified. Wnt3a-signaling activates ROCK, which phosphorylates MYPT1 at Thr853 causing the translocation of MP to the nucleus, where it can form a complex with Nkx2.5. MYPT1 induces exclusion of Nkx2.5 from the nucleus to the cytoplasm in a chromosome region maintenance 1 (Crm1)-dependent pathway. The translocation of Nkx2.5 apparently results in the inhibition of its transcriptional activity [94].

Numerous other MYPT1-interacting proteins such as RNA helicase, protein phosphatase 1B and replication protein A have been identified from human hepatocellular carcinoma cells and the interaction of MYPT1 with the members of the methylosome complex consisting of the protein arginine methyltransferase 5 (PRMT5), MEP50 and PCln was also verified [54]. PRMT5 plays roles in the regulation of transcription, RNA transport and cellular signaling by the generation of the ω-NG, NG-symmetric dimethylarginine. The molecular mechanism how MP regulates gene expression is through the inhibition of PRMT5 activity by its dephosphorylation at the Thr80 residue antagonizing the effect of ROCK. PRMT5 symmetrically dimethylates histone 2AR3 and − 4R3 and shifts the balance from an activating to a suppressive symmetric dimethylarginine mark at these motifs with respect to gene expression. The lack of MYPT1 results in the altered gene expression of 2627 genes involved in cell cycle, tumor morphology and metabolism [54] (Fig. 5).

The fact that the thyroid-hormone stimulation induced the phosphorylation of MYPT1 at Thr853 inhibitory phosphorylation site through the ROCK signaling pathway and caused a decrease of the osteocalcin expression in osteoblasts also support the diverse nuclear role of MP [95]. These lines of evidence support the role of MP in regulating gene expression through MP-mediated dephosphorylation and the identification of target genes opens new perspectives on the function of ROCK/MP signaling at the nuclear level.

2.6. The role of MP in the regulation of endothelial and epithelial barrier function

Vascular endothelium forms a selective permeable barrier between blood and the interstitial space of all organs and participates in the regulation of macromolecule transport and blood cell trafficking through the vessel wall. Endothelial MYPT1 was first cloned from porcine aortic endothelial cells [96]. The involvement of MP in endothelial cell barrier enhancement/protection was directly proved since ROCK-induced phosphorylation of MYPT1 at the inhibitory phosphorylation sites resulted in MP inactivation concomitant with accumulation of phosphorylated MLC20, actin remodeling and cell contraction of the endothelial cells [97].

Nitric oxide (NO) is a soluble gas synthesized from L-arginine in endothelial cells by the endothelial nitric oxide synthase (eNOS). This molecule has a wide range of biological roles such as the modulation of vascular tone by regulating the underlying smooth muscle layer [98]. The activity of eNOS depends on its phosphorylation state as well as its interacting partners and subcellular localization [99]. The major regulatory action is its reversible phosphorylation by PKC or ROCK at its inhibitory Thr497 site. The protein-protein interaction of eNOS and MYPT1 was established, and MYPT1 targeted PP1c to the phosphorylated eNOS-Thr497 substrate. The inhibition of MP, as an eNOS-pThr497-phosphatase suppressed the NO-production and the transendothelial resistance [31] underlying the physiological importance of MP in the maintenance of endothelial integrity and vascular tone.

The expression of full-length MYPT1 isoform was detected in intestinal epithelial cells and it was distributed at the cell-cell contacts and along F-actin [100]. Moreover, airborne particulate induced stress fiber formation in lung epithelial cells via the activation of ROCK and the increased phosphorylation of MYPT1 at Thr853 [101]. Another physiological relevance of MP on barrier function was described in the outer layer of the glomerular filtration barrier consisting of podocytes [102]. These differentiated insulin-dependent epithelial cells prevent protein excretion into the urine by the regulation of their actin cytoskeleton. Insulin increased glomerular filtration barrier permeability through the stimulation of NADPH oxidase that generated reactive oxygen species increasing the dimerization of PKGIα and the phosphorylation of MYPT1 at the inhibitory Thr696 site [103]. These findings support the view of the role of MP in the modulation of barrier integrity through the regulation of the actomyosin complex as well as the production of NO.

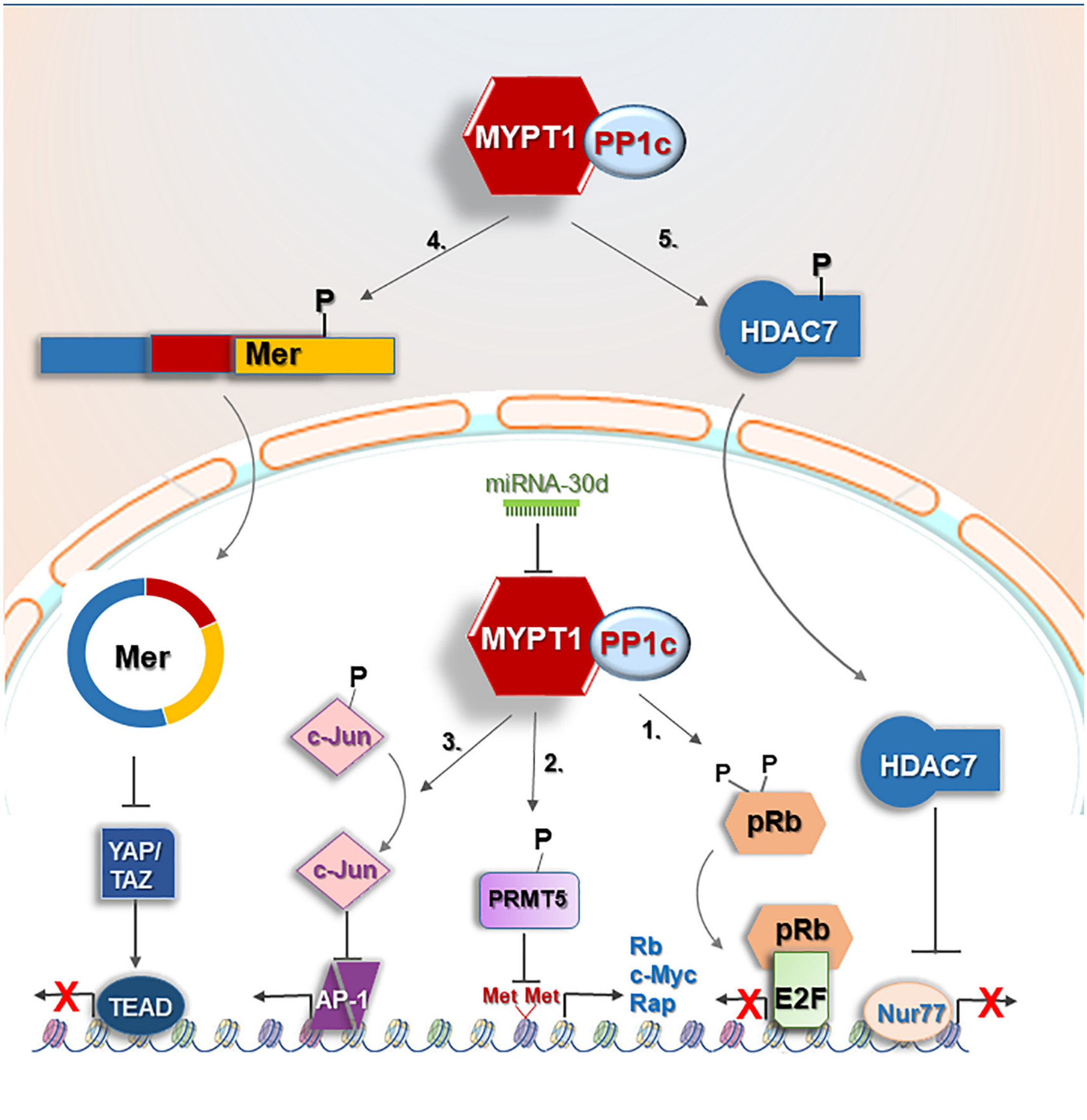

Fig. 5. Anti-tumorigenic signaling pathways regulated by MP.
1. Repression of E2F-mediated cell cycle progression. Activation of pRb by dephosphorylation at Ser807/811 residues causes repression of E2F-mediated cell cycle progression. 2. Activation of the gene expression of tumor-suppressor Rb, Rap and c-Myc. PRMT5 symmetric dimethyltransferase activity is reduced via dephosphorylation catalyzed by MP and results in increased gene expression of tumor suppressors. 3. Inhibition of c-Jun-VEGFA pathway and angiogenesis. MP antagonizes the effect of JNK signaling regulating the phosphorylation of c-Jun at Ser63/73, which increases the AP-1 transcriptional activity enhancing tumorigenic transformation. 4. Activation of anti-mitogenic and tumor suppressor signaling through merlin. Dephosphorylation of merlin at the Ser518 site induces its conformational change, activation and translocation to the nucleus. Active merlin initiates the Hippo pathway (YAP/TAZ/TED) resulting in the degradation of YAP/TAZ oncogenes and inhibition of tumor growth. 5. HDAC7 attaches in the cytoplasm to myosin phosphatase holoenzyme, which dephosphorylates the phospho-Ser residues of HDAC7 and promotes its nuclear redistribution. *E2F1 transcription factor*; *HDAC7*: *histone deacetylase 7*; *Mer*: *merlin*; *miR-30d*: *microRNA-30d*; *MP*: *myosin phosphatase*; *MYPT1*: *myosin targeting subunit 1*; *PP1c*: *protein phosphatase 1 catalytic subunit*; *pRb*: *retinoblastoma protein*; *PRMT5*: *protein arginine methyltransferase 5*; *VEGFA*: *vascular endothelial growth factor A*; *YAP/TAZ*: *transcriptional regulators*.

3. Pathophysiological role of myosin phosphatase

3.1. Anti-tumorigenic action of MP

Several publications have been released recently to elucidate the contribution of the ROCK/MP axis in tumorigenesis beyond their already described role in angiogenesis. Suppression of the RhoA/ROCK signaling pathway with ROCK inhibitors was found to inhibit VEGF-induced angiogenesis and the ability of endothelial cells to organize into capillary networks [104]. It was accompanied by the change of MYPT1 phosphorylation and MP enzyme activity and manifested in the alteration of cytoskeletal reorganization including formation of stress fibers and focal adhesions [105,106]. Although the majority of previous findings suggested the role of MP as a downstream effector of ROCK signaling, more and more results suggest its ROCK-independent action in the inhibition of cancer formation. The unbalanced and constant phosphorylation of MYPT1 at Thr853 was suggested to be related to cancer formation and mutations, in addition altered phosphorylation and expression of MYPT1 were documented in various cancer types [107]. MYPT1 was downregulated in gastric cancer and functional studies revealed that the overexpression of MYPT1 inhibited tumor progression and metastasis formation [108].

MYPT1 phosphorylation and MP activity correlated with the clinical stage of cancers in the case of gastric cancer and in human hepatocellular carcinoma [54], and the higher MYPT1 expression level resulted in longer survival time and also a higher expression of MYPT1 was observed in postoperative patients [108]. c-Myc as a global transcriptional regulator can activate genes involved in cell proliferation and growth but it is also able to repress genes involved in cell cycle and adhesion [109] raising the possibility of a yet-unexplored mechanism used by tumors to suppress differentiation and potentiate aggressiveness. c-Myc expression is significantly lower in HCC than in non-tumor tissues [54]. The overexpression of MYPT1 in gastric cancer cells dramatically improved the expression levels of c-myc and cyclin D suggesting that MYPT1 inhibits cell proliferation and induces cancer cell cycle arrest via cyclin D and c-myc. We found that MYPT1-silencing resulted in a decrease in c-Myc gene and protein expression suggesting again the tumor suppressor effect of this MYPT1 subunit, and presumably MP itself, in HCC (Fig. 5).

Insertional mutations of MYPT1 were also reported to be the drivers of invasive lobular breast cancer (ILC). The majority of ILC is characterized by the functional loss of E-cadherin but mutations of additional genes are also required for the tumorigenesis. Among these genes *mypt*1 was characterized. *Mypt1* gene is located in chromosome 1 q locus, which is frequently gained or amplified in ILC producing truncated MYPT1 subunits. These mutant proteins lacked various regulatory domains altering its scaffolding ability but preserved their PP1c binding feature [110].

Aberrant microRNA 30d (miR30d), a small non-coding RNA molecule, has been associated with cancer progression and its overexpression was observed in prostate cancer cells and in human specimens. MiR-30d enhanced proliferation, migration, invasion, angiogenesis and tumor growth due to downregulation of its target gene, *MYPT1*, which subsequently led to the activation of c-Jun and VEGFA proteins, which are proangiogenic factors. MP signaling activation by inducing MYPT1 expression antagonized the effect of miR-30d, implying that MYPT1 serves as an anti-oncogenic factor in prostate cancer cells and may be a potential drug target [111] (Fig. 5).

Downregulation of *MYPT1* in human hepatocellular carcinoma cells resulted in the upregulation of cancer-related pathways. Above all, MYPT1 inhibitory phosphorylation was significantly increased and in parallel, its nuclear target, PRMT5 enzyme activity (suggested by the elevation of its Thr80 phosphorylation) increased in human hepatocellular carcinoma biopsies and 15 other tumorous cell lines. Hyperactivation of PRMT5 suppressed gene and protein expression of several tumor suppressors such as pRb and RAP1. [54].

In contrast to the previously mentioned anti-oncogenic potential of MP, the Serum Antibodies based SILAC-Immunoprecipitation approach identified MYPT1 as a pancreatic cancer associated antigen upon the application of pancreatic cancer vaccine (GVAX) in phase II clinical trial. Significant MYPT1 expression was detected in malignant compared to normal duct epithelium. Interestingly, tissue microarray revealed significantly higher amount of cytoplasmic and membrane-bound MYPT1 compared to other cancer types [112]. This controversial finding could be resolved by suggesting a pancreatic cancer-specific phenomenon.

Nevertheless, the regulation of protein stability of MYPT1 was also indicated to play a crucial role in cancer formation. The stability of PP1c and MYPT1 is interdependent and knocking down one of the subunits decreases the expression level of the other associated with changes in cell shape and cell survival. [11] Moreover, the protein stability of MYPT1 and its affinity for the ubiquitin-proteasome pathway also play a role in cancer formation through its modification by histone demethylase LSD1 (as it was detailed in the Introduction). LSD1 overexpression and activation in cancer cells induces the destabilization of MYPT1 protein resulting in the activation of pRb phosphorylation and inducing E2F activity [35]. Retinoblastoma protein and merlin also play a key role in cancer formation by modulating gene expression as it was discussed earlier ("Role of MP in the control of cell proliferation and cell division" and Fig. 5). These trends collectively lead to the conclusion that MP is an anti-tumorigenic effector by regulating the gene expression of tumor suppressors in various cancer types. Possible ROCK-independent mechanisms are either through the gene expression regulation of MYPT1 or through the modulation of protein stability of MYPT1 resulting in a decreased level of proteins in cancer cells. However, the regulation of MP holoenzyme by the inhibitory phosphorylation of MYPT1 is another possibility.

3.2. MP in neurodegenerative disorders

MYPT1 was identified as one of the peripheral markers in Huntington's disease (HD) patients and in HD mouse models. The presence of mutant huntingtin caused an increase in the expression level of MYPT1 in human blood leukocytes, postmortem brain as well as in HD mouse tissues. Stimulation of ROCK pathway including MP mainly regulated the cytoskeletal rearrangement and triggered caspase-mediated cell death [113].

Interestingly, MYPT1 was also described as a biomarker in cerebrospinal fluid of patients with Parkinson's disease and its expression was downregulated but its neurological role yet needs to be clarified [114].

3.3. MP in insulin signaling: novel drug target of insulin resistance?

Insulin mediates a wide spectrum of biological processes via the phosphatidylinositide 3 kinase (PI3K) and the MAPK-signaling pathways [115], and the malfunctions of insulin signaling could contribute to the development of insulin resistance, metabolic syndrome and type 2 diabetes [116]. Insulin acts through insulin receptor substrate (IRS-1) that interacts with p85 and Grb2 and they associated with the PI3K and MAPK pathways, respectively, leading to the activation of a number of downstream kinases, such as AKT/PKB, mTOR/raptor, S6K1, GSK3 and PDK1.

In the early 2000s the major function of MP was recognized only as an element of a complex signaling network of insulin that controls the phosphorylation of 20 kDa MLC and VSMC contraction [117]. Insulin-induced IRS-1 tyrosine phosphorylation activates PI3K/AKT signaling and the vasodilatory effect is mediated through NO activating cGKI and resulting in the dephosphorylation of MYPT1 at the inhibitory phosphorylation site. The phosphorylation of MYPT1 was negatively regulated by AKT (protein kinase B) phosphorylation in response to insulin, therefore, phosphorylation and activation of AKT, downstream of IRS-1 and PI3-K, were impaired in diabetic VSMC [118]. Besides that, insulin triggered serotonin-induced contraction through the IR/PI3K/PDK1 pathway and PDK was suggested to directly phosphorylate MYPT1 in a ROCK-independent manner [119]. Interestingly, the hyper-reactivity of diabetic patients' VSMC was the consequence both the enhanced phosphorylation of MP but also the decreased expression level of MYPT1 [120].

A relatively new component of the MYPT1-PP1cδ complex, myosin phosphatase-Rho interacting protein (M-RIP) also plays a role in the regulation of MP in insulin signaling. M-RIP is a smooth muscle protein [121] that binds to actin filaments through its amino-terminal domain and acts as a scaffold to RhoA/ROCK to regulate MP on the actomyosin complex [122]. Insulin stimulation initiated the binding of M-RIP to MYPT1 resulting in an insulin-induced vasodilation in vascular smooth muscle [123] (Fig. 6).

In skeletal muscle, insulin stimulation leads to the activation of PI3K/PDK1/Akt signaling initiating mTOR that required for the interaction of MYPT1 with IRS1. MP dephosphorylate IRS1 at the phospo-Ser residues counterbalancing the effect of the insulin-activated protein kinases. MYPT1 was identified as a novel endogenous insulin-stimulated interaction partner of IRS-1 in L6 myotubes [124]. Insulin initiated an interaction between the IRS1/p85 complex and MYPT1, and it was suggested to be dependent on the activation of PI3K, PDK1, AKT/PKB and mTOR/raptor (mTORC1) but not MAPK signaling. Based on

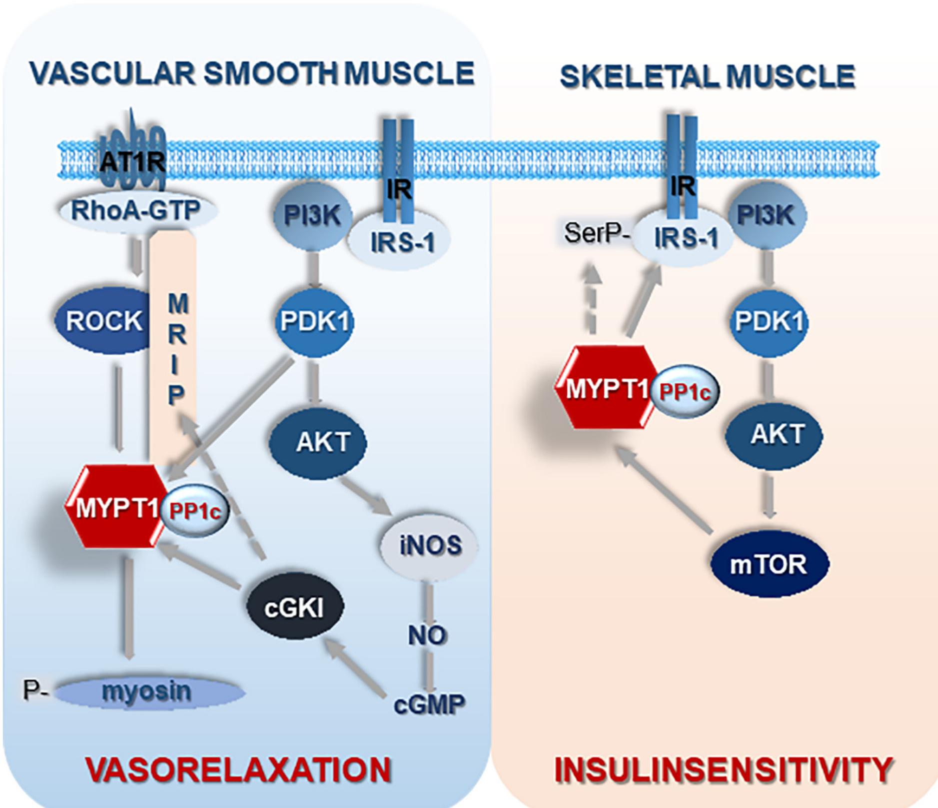

Fig. 6. Hypothetic function of MP in insulin signaling.
Insulin-induced IRS1 tyrosine phosphorylation activates PI3K/Akt signaling and the vasodilatory effect is mediated through NO activating cGKI and resulting in the dephosphorylation of MYPT1 at the inhibitory phosphorylation site. Upon insulin stimulation, M-RIP binds to MYPT1 causing vasodilatation in vascular smooth muscle. In skeletal muscle, insulin stimulation leads to the activation of PI3K/PDK1/Akt signaling initiating mTOR that is required for the interaction of MYPT1 with IRS1. MP dephosphorylates IRS1 at the phospho-Ser residues counterbalancing the effect of the insulin-activated protein kinases.
cGKI: *cyclic guanosine-3′,5′-monophosphate-dependent protein kinase*; *iNOS*: *inducible nitric oxide synthase*; *IR*: *insulin receptor*; *IRS1*: *insulin-receptor substrate 1*; *mTOR*: *mammalian target of rapamycin*; *MRIP*: *myosin phosphatase-Rho interacting protein*; *PDK1*: *phosphoinositide-dependent kinase-1*; *PI3K*: *phosphatidylinositide 3 kinase*.

these results, the hypothetic function of MP is the dephosphorylation of IRS-1 to counterbalance the effect of insulin-activated Ser/Thr kinases, resulting in balanced phosphorylation of IRS-1 on Ser and Thr residues and proper insulin action through IRS1 (Fig. 6). Insulin stimulation led to increased interaction between IRS-1 and MYPT1/PP1cδ, and to the greater ability of MP to dephosphorylate IRS-1 [125]. These lines of evidence were supported by quantitative phosphoproteomics assay using *MYPT1* knock down L6 skeletal muscle cell lines and 295 phosphoproteins were documented to be related in insulin signaling to mTOR or RhoA signaling [126].

4. Conclusions

Recent publications highlighted the diverse and unexpected functions of myosin phosphatase, the well-characterized enzyme in smooth muscle contractility. It has already been documented that the MYPT1 regulatory subunit of MP is ubiquitously expressed and it exhibits diverse subcellular localization in eukaryotic cells. The fact that besides the cytoplasm and cytoskeleton MYPT1 and PP1c both localized in nuclear, mitochondrial and plasma membrane fractions of the cells, and that the majority of identified proteins in MYPT1-interactomes are not relevant to cytoskeletal regulation suggest that MP is more than the regulator of myosin light chain dephosphorylation only.

One key observation highlighted in this review is that MYPT1 functions not simply as a substrate targeting subunit in MP, but it could also serve as a scaffold protein or provides a nucleocytoplasmic shuttle mechanism for other proteins and PP1c activation/inhibition are not always coupled with these events. On the other hand, MP has been considered for decades as the downstream target of the RhoA /ROCK signaling, nevertheless its therapeutic potential has been deeply ignored. Several reports have now described the ROCK-independent action and regulation of MP, therefore this review has intended to focus on the importance of MP in distinct signaling mechanisms.

Based on the findings on the role of MP in distinct pathways, the modulation of MP by designing MP holoenzyme-specific activator(s) provides plausible way of repressing tumorigenesis or initiating insulin sensitivity, while enhancing dephosphorylation of crucial downstream molecules such as pRb, PRMT5 or PLK may also provide a novel therapeutic approach. The combination of genetic and pharmacological tools will allow exploration of these relationships and it may contribute to a more complete understanding of the molecular events regulated by dephosphorylation processes.

Transparency document

The Transparency document associated with this article can be found, in online version.

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Conflict of interest

The authors have declared that no competing interests exist.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbamcr.2018.07.023.

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